

Cochin university college of engineering kuttanad

Cochin university of science and technology

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

23-200-0104B BASIC ELECTRICAL ENGINEERING LAB

23-20-108B ELECTRICAL ENGINEERING LABORATORY

Course Outcomes

1. Identify & choose appropriate apparatus for ON-OFF Control, protection diagnosis & instrumentation of a typical LV electrical appliance/circuit.
2. Familiarize with various types of electric motors and conventional & smart electrical systems.
3. Familiarize with the electrical characteristics of common appliances & solar panel.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3								2	2						
CO2	3								2	2						
CO3	3								2	2						

Details of Experiments

1. Familiarization with various electrical apparatus like switches, relays, smart plugs, smart switches, conventional to smart switch converters, AC & DC Voltmeter, AC & DC Ammeter, Multimeter, Wattmeter, Energy meter, fuse, MCB, Isolator, RCB, ELCB, RCBO
2. Verification of Ohm's Law & Kirchoff's Laws for both dc & ac circuits.
3. Domestic Wiring Circuits with one way/two-way switches & plug point.
4. Experimental determination of V-I characteristics, MPP & predetermination of operating point of a solar panel for resistive load.
5. Measurement of Current, Power, Power Factor & Energy of: a. A single-phase circuit with known parameters b. Various domestic/daily use appliances like LED Lamps, LED Tubes, Ceiling fans, Laptop, LED Display, PC+LED Display.
6. Experimental/Simulation based study of an RLC series Circuit under resonant & non resonant conditions. 7. Experimental determination of frequency Response of Circuits

8. Familiarization with various types of Electrical Machines

9. Experiment on Automatic Street Lighting System.

10. Experiments on Home Automation.

GENERAL LAB RULES

All students in the lab are expected to adhere to the following guidelines.

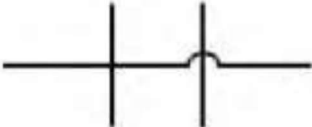
- ☐ ☐ The students are supposed to come in proper workshop uniform dress. Wearing shoes in the lab is compulsory.
- ☐ ☐ To successfully complete the experiments in one lab period, you must come prepared to the laboratory.
- ☐ ☐ Return the components to the lab staff when you are finished with them.
- ☐ ☐ Before leaving the lab, turn off the main power switch to the lab bench.
- ☐ ☐ Keep your work area neat. Have only books and other materials that are needed to conduct the experiment in the laboratory.

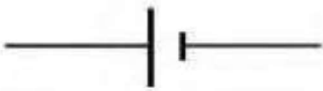
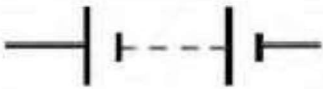
ELECTRICAL SAFETY PRINCIPLES

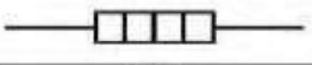
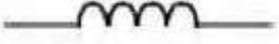
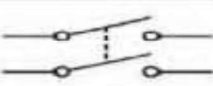
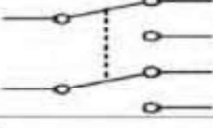
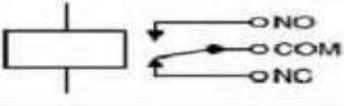
When planning and performing work on electrical systems and equipment, keep these principles in mind:

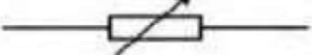
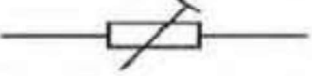
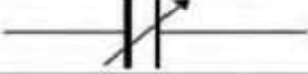
- ☐ ☐ Understand the procedure completely before starting the work.
 - ☐ ☐ Use good quality footwear/shoes in order to provide maximum resistance.
 - ☐ ☐ Never energize any circuit unless you are sure that no one is working on the circuit. Give electric supply to the wiring system only after thorough verification.
 - ☐ ☐ Before replacing a blown fuse always remember to put the switch off.
 - ☐ ☐ Do not touch switch boards, main switches, holder points etc with wet hands.
 - ☐ ☐ Do not use broken switches, sockets or plug.
 - ☐ ☐ Use non-conductive tools whenever possible.
- ✓ Before putting the plug pins in socket put off the plug switch and disconnect the plug by pulling the plug pin and not by pulling cable.
- ☐ ☐ Take utmost care while handling lamps, lamp holders, switches etc., because these materials are brittle. Before beginning work, tie back long hair, and roll up loose sleeves

SYMBOLS

COMPONENT	CIRCUIT SYMBOL	DESCRIPTION
WIRE CONNECTIONS		
Wire		To pass current very easily from one part of a circuit to another.
Wires joined		A 'blob' should be drawn where wires are connected (joined), but it is sometimes omitted. Wires connected at 'crossroads' should be staggered slightly to form two T-junctions, as shown.
Wires not joined		In complex diagrams it is often necessary to draw wires crossing even though they are not connected. (prefer the 'bridge' symbol shown on the right because the simple crossing on the left may be misread as a join where you have forgotten to add a 'blob'!)

POWER SUPPLIES		
Cell		Supplies electrical energy. The larger terminal (on the left) is positive (+). A single cell is often called a battery, but strictly a battery is two or more cells joined together.
Battery		Supplies electrical energy. A battery is more than one cell. The larger terminal (on the left) is positive (+). The smaller terminal (on the right) is negative (-).
DC supply		Supplies electrical energy. DC = Direct Current, always flowing in one direction.
AC supply		Supplies electrical energy. AC = Alternating Current, continually changing direction.
Fuse		A safety device which will 'blow' (melt) if the current flowing through it exceeds a specified value.
Transformer		Two coils of wire linked by an iron core. Transformers are used to step up (increase) and step down (decrease) AC voltages. Energy is transferred between the coils by the magnetic field in the core. There is no electrical connection between the coils.
Earth (Ground)		A connection to earth. For many electronic circuits this is the 0V (zero volts) of the power supply, but for mains electricity and some radio circuits it really means the earth. It is also known as ground.

OUTPUT DEVICES: LAMPS, HEATER, MOTOR, etc.		
Lamp (lighting)		A transducer which converts electrical energy to light. This symbol is used for a lamp providing illumination, for example a car headlamp or torch bulb.
Lamp (indicator)		A transducer which converts electrical energy to light. This symbol is used for a lamp which is an indicator, for example a warning light on a car dashboard.
Heater		A transducer which converts electrical energy to heat.
Motor		A transducer which converts electrical energy to kinetic energy (motion).
Bell		A transducer which converts electrical energy to sound.
Buzzer		A transducer which converts electrical energy to sound.
Inductor (Coil, Solenoid)		A coil of wire which creates a magnetic field when current passes through it. It may have an iron core inside the coil. It can be used as a transducer converting electrical energy to mechanical energy by pulling on something.
Switches		
Push Switch (push-to-make)		A push switch allows current to flow only when the button is pressed. This is the switch used to operate a doorbell.
Push-to-Break Switch		This type of push switch is normally closed (on), it is open (off) only when the button is pressed.
On-Off Switch (SPST)		SPST = Single Pole Single Throw Switch. An on-off switch allows current to flow only when it is in the closed (on) position.
2-way Switch (SPDT)		SPDT = Single Pole Double Throw Switch. A 2-way changeover switch directs the flow of current to one of two routes according to its position. Some SPDT switches have a central off position and are described as 'on-off-on'.
Dual On-Off Switch (DPST)		DPST = Double Pole, Single Throw Switch. A dual on-off switch which is often used to switch mains electricity because it can isolate both the live and neutral connections.
Reversing Switch (DPDT)		DPDT = Double Pole, Double Throw Switch. This switch can be wired up as a reversing switch for a motor. Some DPDT switches have a central off position.
Relay		An electrically operated switch, for example a 9V battery circuit connected to the coil can switch a 230V AC mains circuit. NO = Normally Open, COM = Common, NC = Normally Closed.

Resistors		
Resistor		A resistor restricts the flow of current, for example to limit the current passing through an LED. A resistor is used with a capacitor in a timing circuit. (Some publications also use the old resistor symbol: )
Variable Resistor (Rheostat)		This type of variable resistor with 2 contacts (a rheostat) is usually used to control current. Examples include: adjusting lamp brightness, adjusting motor speed, and adjusting the rate of flow of charge into a capacitor in a timing circuit.
Variable Resistor (Potentiometer)		This type of variable resistor with 3 contacts (a potentiometer) is usually used to control voltage. It can be used like this as a transducer converting position (angle of the control spindle) to an electrical signal.
Variable Resistor (Preset)		This type of variable resistor (a preset) is operated with a small screwdriver or similar tool. It is designed to be set when the circuit is made and then left without further adjustment. Presets are cheaper than normal variable resistors so they are often used in projects to reduce the cost.
CAPACITORS		
Capacitor		A capacitor stores electric charge. A capacitor is used with a resistor in a timing circuit. It can also be used as a filter, to block DC signals but pass AC signals.
Capacitor polarized		A capacitor stores electric charge. This type must be connected the correct way round. A capacitor is used with a resistor in a timing circuit. It can also be used as a filter, to block DC signals but pass AC signals.
Variable Capacitor		A variable capacitor is used in a radio tuner.
Trimmer Capacitor		This type of variable capacitor (a trimmer) is operated with a small screwdriver or similar tool. It is designed to be set when the circuit is made and then left without further adjustment.
DIODES		
Diode		A device which only allows current to flow in one direction.
LED Light Emitting Diode		A transducer which converts electrical energy to light.
Zener Diode		A special diode which is used to maintain a fixed voltage across its terminals.

EXPERIMENT NO. 1**PLUG POINT WIRING****AIM :-**

- 1) To wire up a circuit to obtain one plug point and one lamp controlled by independent switches
- 2) To estimate the materials required and set up the above circuit using PVC conduit system

TOOLS REQUIRED:-

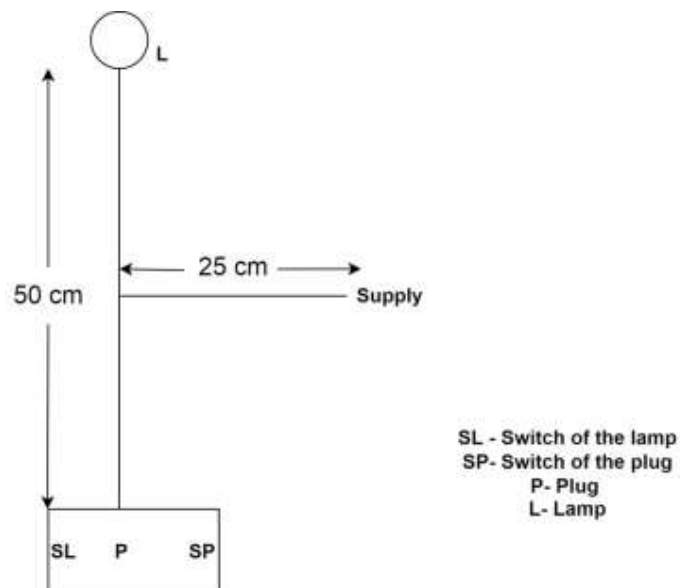
Screwdriver, cutting pliers, tester, poker, stripper, ball peen hammer.

DESIGN:

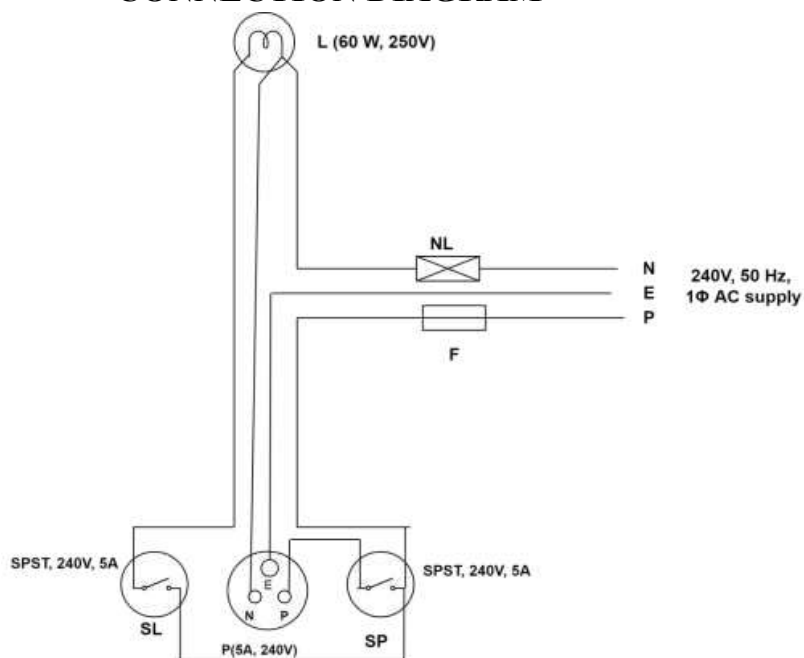
Power =

Load current = Power / 230

Fuse Wire rating = 1.5 * Load current

LAY OUT

CONNECTION DIAGRAM



OBSERVATION TABLE

SL	SP	L	P
ON	ON	BRIGHT	LIVE
OFF	OFF	DARK	DEAD
OFF	ON	DARK	LIVE
ON	OFF	BRIGHT	DEAD

MATERIALS REQUIRED

SL No:	Materials required	Specification	Quantity
1			

PROCEDURE :-

The dimensional layout is drawn on the board using chalk. Conduits are placed on the board along the layout drawn and fixed using saddles. Lay down the wires through the conduit as per the circuit diagram. Round block, switch and holder are fixed on the board after giving connections as given in fig. Neutral link and fuse unit are also inserted in the circuit and connections are given. The wired circuit is tested by providing 240 V, 50Hz, single phase supply. The lamp glows according to switching conditions, the wired circuit is found correct. If the lamp does not glow or fuse is blown out, the circuit has to be checked for any error.

RESULT:-

EXPERIMENT NO. 4**STAIRCASE WIRING****AIM :-**

- 1) To wire up a circuit to control one lamp using two switch.
- 2) To estimate the materials required and set up the above circuit using PVC conduit system.

TOOLS REQUIRED:-

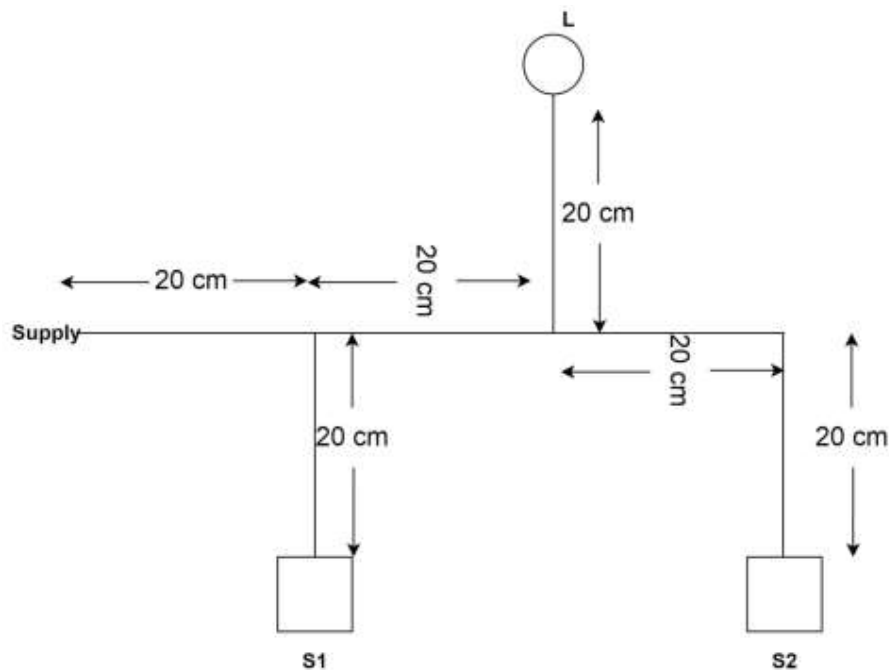
Screwdriver, cutting pliers, tester, poker, stripper, ball peen hammer.

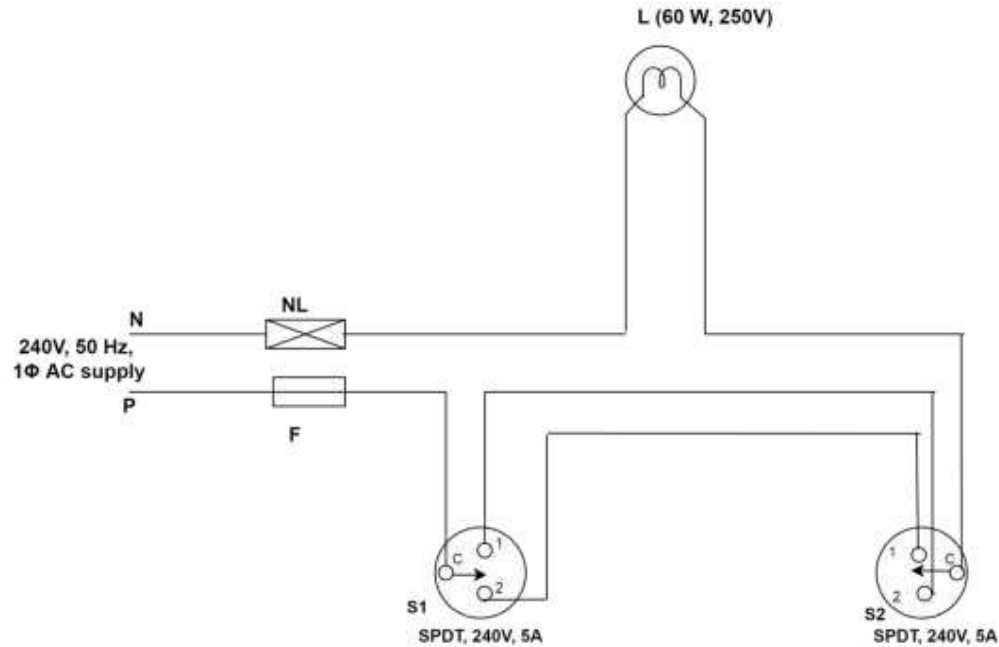
DESIGN:

Power =

Load current = Power / 230

Fuse Wire rating = $1.5 \times$ Load current

LAY OUT

CONNECTION DIAGRAM**OBSERVATION TABLE**

S1	S2	L
UP	UP	OFF
UP	DOWN	BRIGHT
DOWN	UP	BRIGHT
DOWN	DOWN	OFF

MATERIALS REQUIRED

Sl no	Materials required	Specification	Quantity
1			
2			

PROCEDURE :-

The dimensional layout is drawn on the board using chalk .Conduits are placed on the board along the layout drawn and fixed using saddles. Lay down the wires through the conduit as per the circuit diagram. Roundblock, switch and holder are fixed on the board after giving connections as given in fig. Neutral link and fuse unit are also inserted in the circuit and connections are given. The

wired circuit is tested by providing 240 V, 50Hz, single phase supply. The lamp glows according to switching conditions, the wired circuit is found correct. If the lamp does not glow or fuse is blown out, the circuit has to be checked for any error.

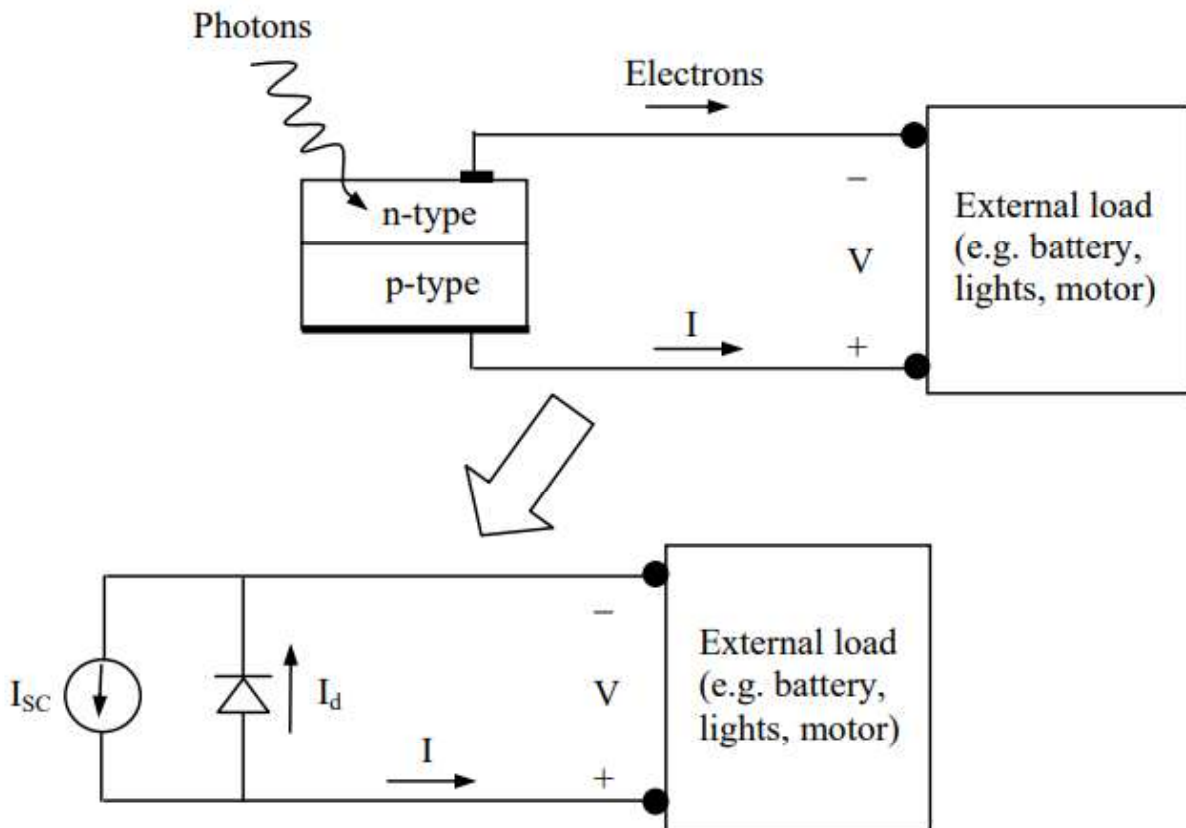
RESULT

V-I Characteristics & MPP of Solar Panel

Aim: To sketch the V-I Characteristics of the given Solar Panel, to find its Maximum Power Point and to predetermine the V & I when the Panel is terminated in a 20Ω resistance and.

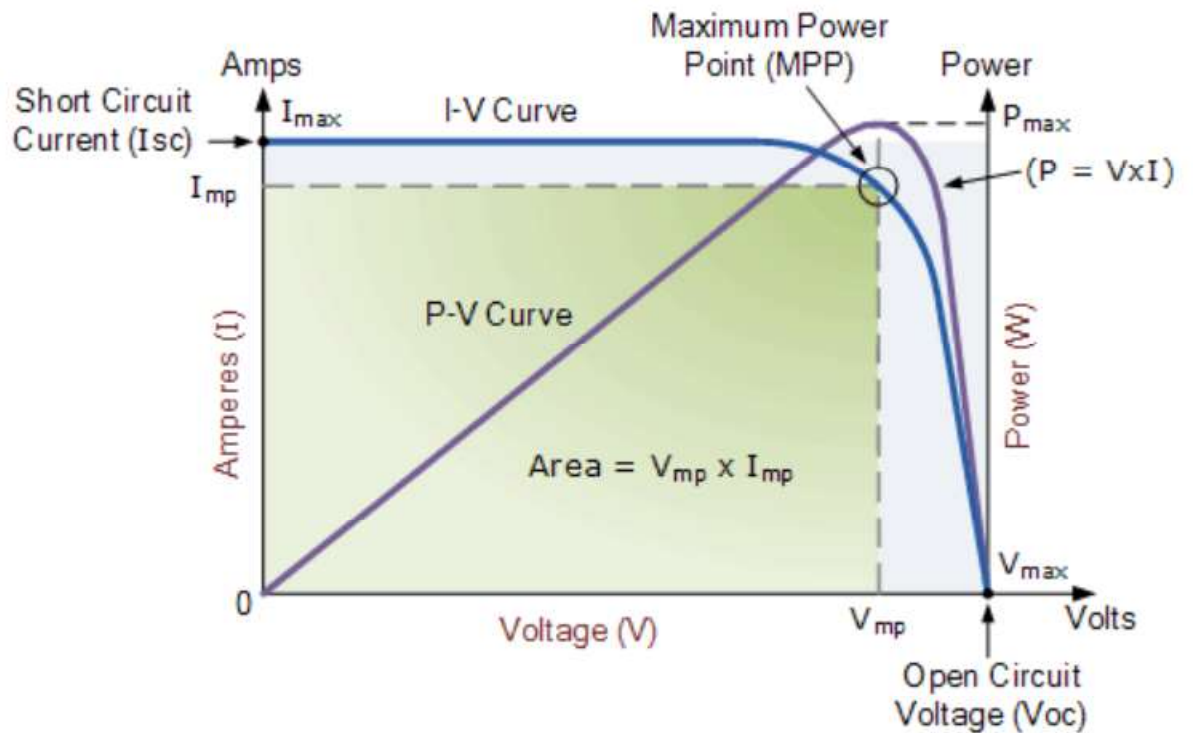
Theory

Incident sunlight can be converted into electricity by photovoltaic conversion using a solar panel. A solar panel consists of individual cells that are large-area semiconductor diodes, constructed so that light can penetrate into the region of the p-n junction. The junction formed between the n-type silicon wafer and the p-type surface layer governs the diode characteristics as well as the photovoltaic effect. Light is absorbed in the silicon, generating both excess holes and electrons. These excess charges can flow through an external circuit to produce power.

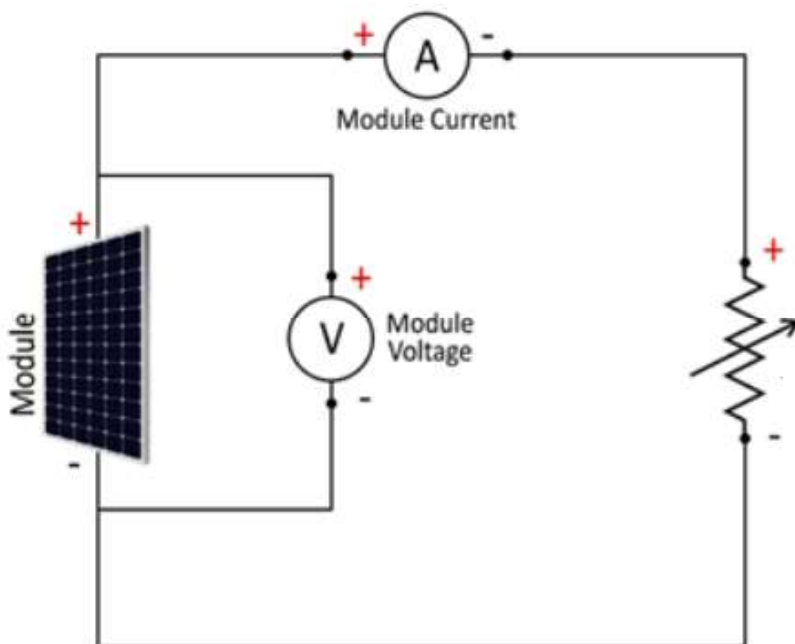


The approximate equivalent circuit of a solar cell consists of a current source & a diode in parallel. If the solar cell is open circuited, then all of the I_{sc} flows through the diode and produces an open circuit voltage V_{oc} of about 0.5-0.6V. If the solar cell is short circuited, then no current flows through the diode, and all of the short-circuit current I_{sc} flows through the short circuit. Practical solar panels consist of multiple solar cells in series-parallel combinations. The

typical V-I curve of a solar panel for a particular isolation level & the corresponding P-V curve is shown below.



Circuit



Procedure

- Connections are made as shown in the circuit diagram.
- By adjusting the Rheostat, a set of readings are taken & tabulated.
- The data is copied to a graph to get the V-I Characteristics of the given Solar Panel.
- The terminal voltage V vs output Power P curve of the Panel is plotted. From the plot, the MPP is determined.
- V-I curve corresponding to 20Ω resistance is plotted in the same graph as the V-I curve of the Solar Panel. The point of intersection of the two plots gives the V & I for 20Ω resistance termination.

Tabulation

S.N	Terminal Voltage, V (V)	Load Current, I (A)	Power, P (W)
1.			
2.			

Open Circuit Voltage =.....(V)

Short Circuit Current =.....(A)

Maximum Power =....(W)

Voltage corresponding to MPP =...(V)

Current corresponding to MPP = ...(A)

Result

The experiment has been completed successfully.

Introduction to Circuit Simulation

Aim

1. To study about the condition for balance & the effect of various circuit elements upon the condition for balance of a Wheatstone Bridge by simulation.
2. To compute current through & voltage across every passive element of a multi source-multi loop circuit and to verify the results by simulation.

Theory & Procedure of Wheatstone Bridge Experiment

Theory

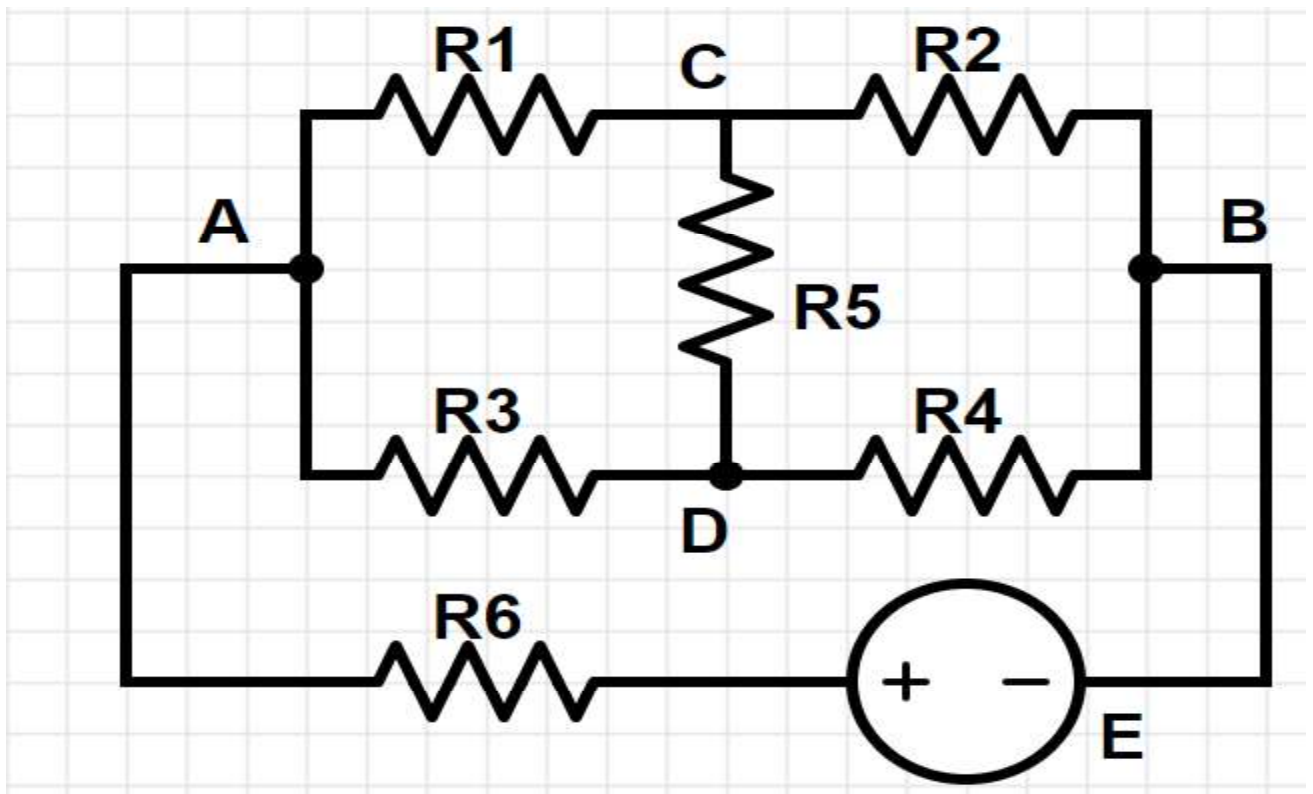
The Wheatstone Bridge is said to be in balance when current through/voltage across the central branch of the bridge is zero. Hence for a balanced bridge,

$$I_{CD} = 0, V_{CD} = 0$$

$$V_{AC} = V_{AB} \left(\frac{R_1}{R_1 + R_2} \right) = V_{AD} = V_{AB} \left(\frac{R_3}{R_3 + R_4} \right)$$

$$V_{CB} = V_{AB} \left(\frac{R_2}{R_1 + R_2} \right) = V_{DB} = V_{AB} \left(\frac{R_4}{R_3 + R_4} \right)$$

Therefore the condition for balance is that $\frac{R_1}{R_2} = \frac{R_3}{R_4}$



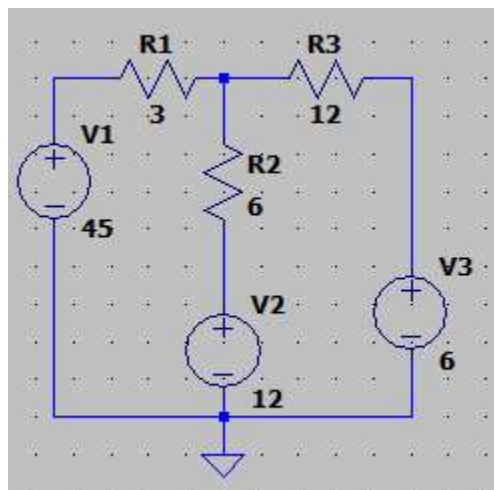
Procedure

1. The model of a Wheatstone Bridge is developed in LTspice.
2. The current through the central branch of the bridge circuit is recorded for various values of the circuit elements.
3. The effect of various circuit elements upon the condition for balance is inferred from the tabulated data.

Tabulation

S.N	E(V)	R1(Ω)	R2(Ω)	R3(Ω)	R4(Ω)	R5(Ω)	R6(Ω)	I _{cd} (A)	Whether (R1/R2)=(R3/R4) or not (Yes/No)
1.									
2.									
3.									
4.									
5.									
6.									

Procedure for Experiment on Multi Source-Multi Loop Circuit



1. The current through & voltage across every element of the given multi source-multi loop circuit is computed.
2. The results are verified by running the model of the circuit in LTspice.

Tabulation

Calculated Values						Observed Values					
I _{3Ω} (A)	V _{3Ω} (V)	I _{6Ω} (A)	V _{6Ω} (V)	I _{12Ω} (A)	V _{12Ω} (V)	I _{3Ω} (A)	V _{3Ω} (V)	I _{6Ω} (A)	V _{6Ω} (V)	I _{12Ω} (A)	V _{12Ω} (V)

Result

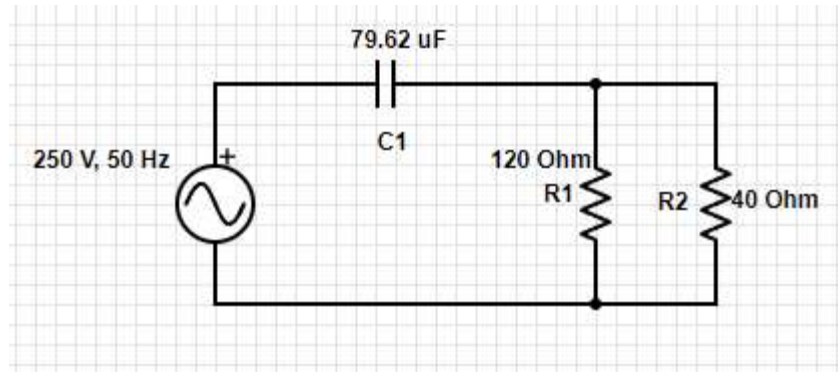
The experiment has been completed successfully.

Verification of Basic Circuit Laws-II

Aim

1. To compute current through & voltage across every passive element of the given circuit and to verify the results by simulation.
2. To experimentally verify Ohm's Law & Kirchhoff's Laws for AC circuits.

Procedure



1. The current through & voltage across every element of the given circuit is computed as phasors.
2. The results are verified by running the model of the circuit in LTspice.
3. Ohm's Law, KCL & KVL are verified by using the data from the simulation experiment.

Tabulation

Observed Values					Calculated Values							
$I_C(A)$	$I_{R1}(A)$	$I_{R2}(A)$	$V_C(V)$	$V_{R1}=V_{R2}(V)$	$I_C(A)$	$I_{R1}(A)$	$I_{R2}(A)$	$V_C(V)$	$V_{R1}=V_{R2}(V)$	$I_{R1}+I_{R2}(A)$	$V_C+V_{R2}(V)$	$Z=V/I_C(\Omega)$

Sample Calculation

Verify that $I_C=I_{R1}+I_{R2}$ & $V=V_C+V_{R2}$

....

.....

Experimental value of Effective impedance, $Z_A=.....(\Omega)$

Theoretical Value of Effective Impedance, $Z_B=-jX_C+(R1//R2)=.....(\Omega)$, $X_C=1/\omega C$

Verify that $Z_A=Z_B$

Result

The experiment has been completed successfully.

(Theoretical Background required for completing the Experiment: Basics of AC Circuits, Phasor Algebra)

Relays & Simple Automation Circuits

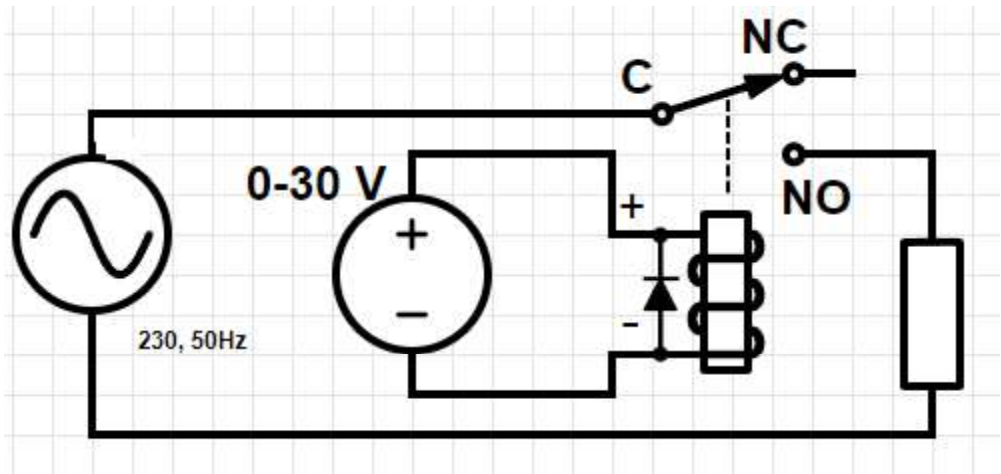
Aim

- 1. To sketch the transfer characteristics of the given relay circuit.
- 2. To design and test an interface circuit for controlling a 12V relay from a 5V source.

Procedure

Relay Transfer Characteristics

- 1. The terminals of the relay are identified with a multimeter.
- 2. Connections are made as shown in the figure.



- 3. By adjusting the voltage applied across the relay coil, the pull-in voltage & drop out voltage of the relay are recorded. The data is used for sketching the variation of the voltages of the normally closed & normally open contacts of the relay with respect to the voltage applied across the relay coil.

Tabulation

Relay Details	Pull-in Voltage (V)	Drop Out Voltage (V)	Hysteresis Width (V)
12V SPDT Relay			

Relay Driver

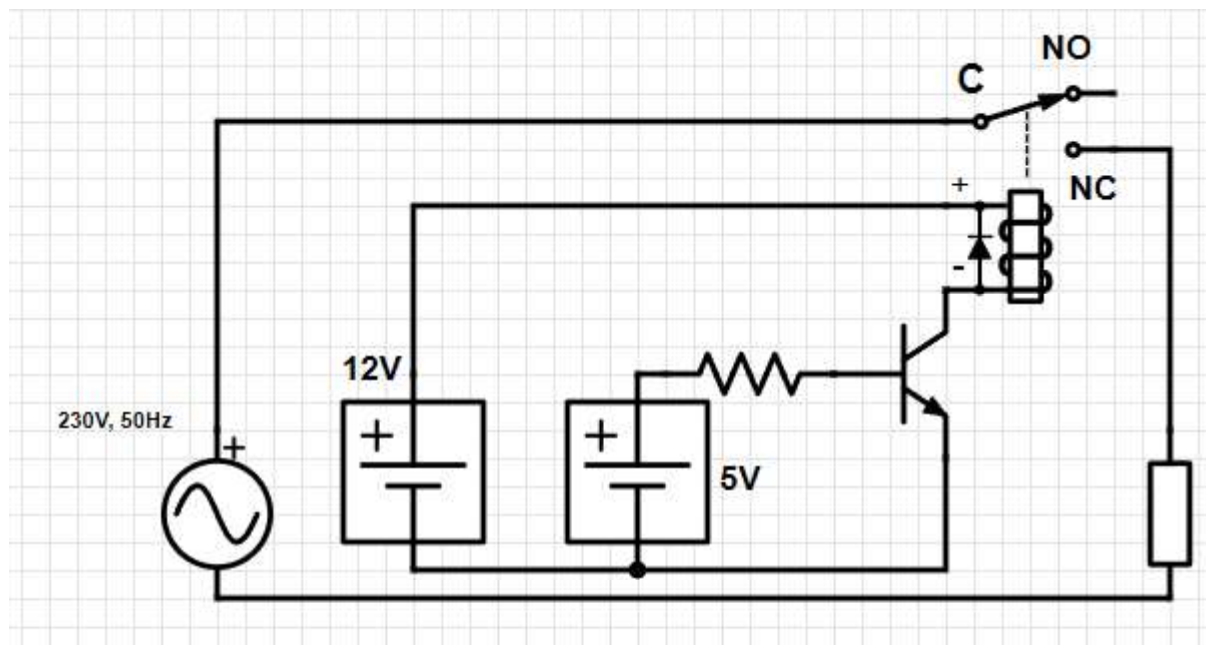
- 1. The resistance of the given relay coil is determined using a multimeter & rated value of coil current is computed.
- 2. Choose suitable transistor & components for the relay driver circuit based upon the following equations:

$$I_{coil} = V_{coil} / R_{coil} , I_{Bmin} = I_{coil} / \beta_{min}$$

Choosing an overdrive factor of 2.5, $I_B = 2.5 I_{Bmin}$

$$R_B = (V_{BB} - V_{BEsat}) / I_B$$

- 3. Wire up the complete circuit & testify the circuit.



Result

The experiment has been completed successfully.

(Reference: <https://en.wikipedia.org/wiki/Relay>)

Power Measurement

Aim

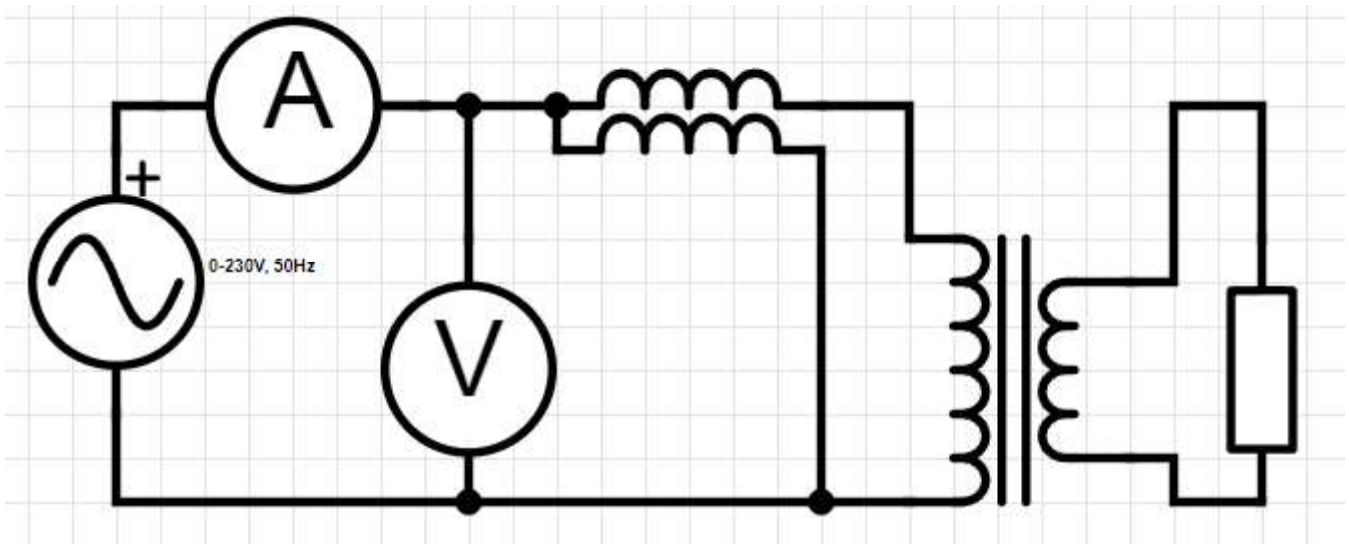
To measure the current drawn, power consumed & power factor of the following single phase loads:

1. A resistive load fed through a transformer
2. A PC with monitor
3. A PC Monitor
4. A pedestal fan

Procedure

Measurements on a transformer with resistive load in the Secondary

1. Connections are made as shown in the circuit diagram.
2. By varying the source voltage, the readings of the meters are tabulated.
3. Based upon the data, computations are performed for finding power factor, reactive power & hourly energy consumption in each case.

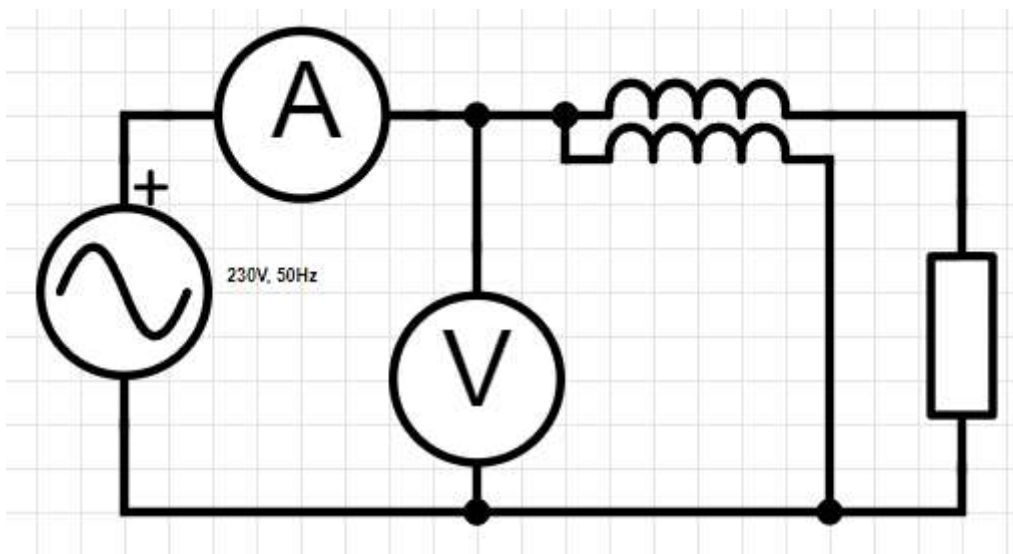


Tabulation

S.N	Voltage (V)	Current (A)	Active Power (W)	Power Factor	Reactive Power (VAR)	Hourly Energy Consumption (J)	Hourly Energy Consumption (kWh)

Measurements on PC with Monitor, PC without monitor & Pedestal Fan

1. Connections are made as shown in the circuit diagram for each load & meter readings are taken.



Tabulation

Item/Particulars	Voltage (V)	Current (A)	Active Power (W)	Power Factor	Reactive Power (VAR)	Hourly Energy Consumption (J)
PC With LCD Monitor						
PC Without Monitor						
Pedestal Fan (Low Speed)						
Pedestal Fan (Medium Speed)						
Pedestal Fan (High Speed)						

Sample Calculations (Set No:.....)

Result

The experiment has been completed successfully.

Frequency Response of Circuits

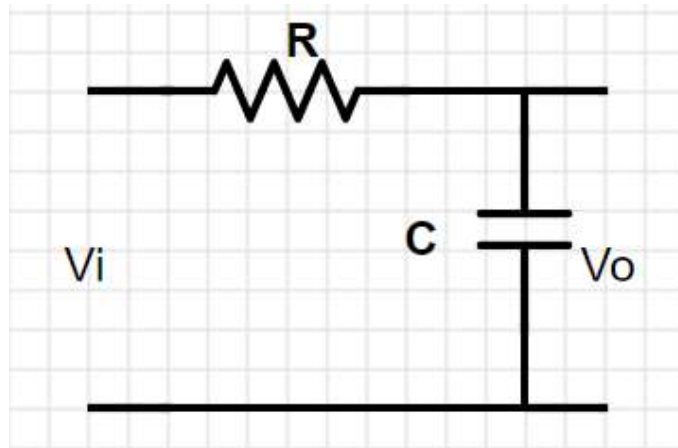
Aim

1. To sketch the frequency response of a first order RC LPF by simulation, determine its 3-dB cut off frequency & validate the frequency response data at a few sample points by calculation and time domain simulation.
2. To sketch the frequency response of a first order RC HPF by simulation, determine its 3-dB cut off frequency & validate the frequency response data at a few sample points by calculation and time domain simulation.

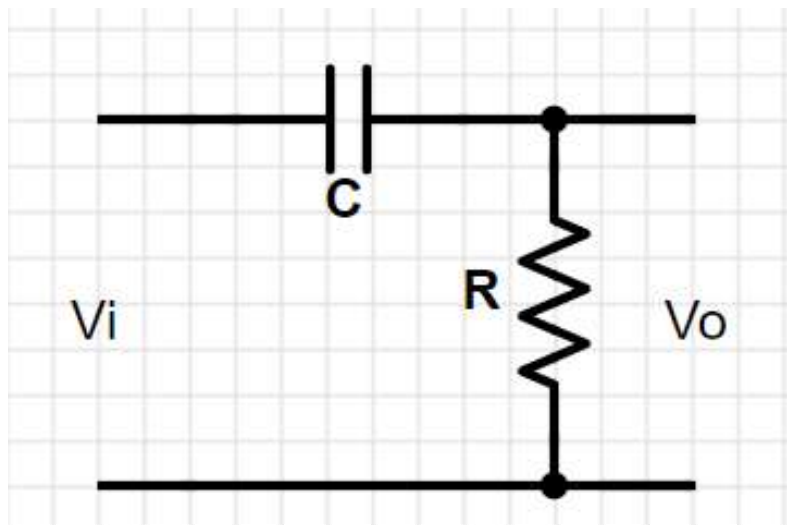
Procedure

1. Model of the first order RC LPF/HPF is developed in LTspice.
2. By running the model in AC Analysis Mode, the frequency response of the circuit is observed & 3-dB cut off frequency of the circuit is determined.
3. The frequency response data is tabulated at a few sample points.
4. The data is validated by calculation & time domain simulation.
5. Observe the input and output waveforms of the filter for an input of $v = v_1 + v_2 + v_3$ where v_1 is a DC signal, v_2 is a sinusoid in the passband of the filter & v_3 is a sinusoid in the stopband of the filter.

Circuit of First Order LPF



Circuit of First Order HPF



Tabulation for LPF

S.N	Freequency (Hz)	Gain Magnitude (dB)			Phase (Degrees)		
		AC Analysis Mode of Simulation	Transient Analysis Mode of Simulation	Calculation	AC Analysis Mode of Simulation	Transient Analysis Mode of Simulation	Calculation

Tabulation for HPF

S.N	Freequency (Hz)	Gain Magnitude (dB)			Phase (Degrees)		
		AC Analysis Mode of Simulation	Transient Analysis Mode of Simulation	Calculation	AC Analysis Mode of Simulation	Transient Analysis Mode of Simulation	Calculation

Cutoff Frequencies of the Filters

Particulars	Experimental Value (Hz)	Theoretical Value (Hz)
3-dB cut off frequency of the LPF		
3-dB cut off frequency of the HPF		

Sample Calculations (Set No:.....)

Result

The experiment has been completed successfully.

(Theoretical Background required for completing the Experiment: Sinsoidal Steady State Response/Frequency Response Function, Frequency Response Plots)

RLC Series Circuit

Aim

1. To study the sinusoidal steady state response of an RLC series circuit.

Procedure

1. Model of the circuit is developed in LTspice.
2. By running the model in the **AC Analysis Mode**, the current vs frequency plot of the circuit is observed.
3. From the current vs frequency plot, resonant frequency, lower half power frequency, upper half power frequency, bandwidth & quality factor of the circuit are determined.
4. The results are verified by calculations.
5. By running the model in the **Transient Analysis Mode** at resonant frequency, current through the circuit, voltage across all the elements of the circuit & voltage magnification factor (M) are determined and the results are verified by calculations.

Tabulations

AC Analysis Mode Simulation

Observed Values					Calculated Values				
f_r (Hz)	f_L (Hz)	f_H (Hz)	BW (Hz)	$Q=f_r/BW$	f_r (Hz)	f_L (Hz)	f_H (Hz)	BW (Hz)	$Q=f_r/BW$

Transient Analysis Mode Simulation

Observed Values						Calculated Values				
V (V)	I(A)	V_R (V)	V_L (V)	V_C (V)	M	I(A)	V_R (V)	V_L (V)	V_C (V)	$M=\omega_r L/R=Q$

Calculations

....

.....

Result

The experiment has been completed successfully.

(Theoretical Background required for completing the Experiment: Sinusoidal Steady State Response of RLC Series Circuit)

Home Automation

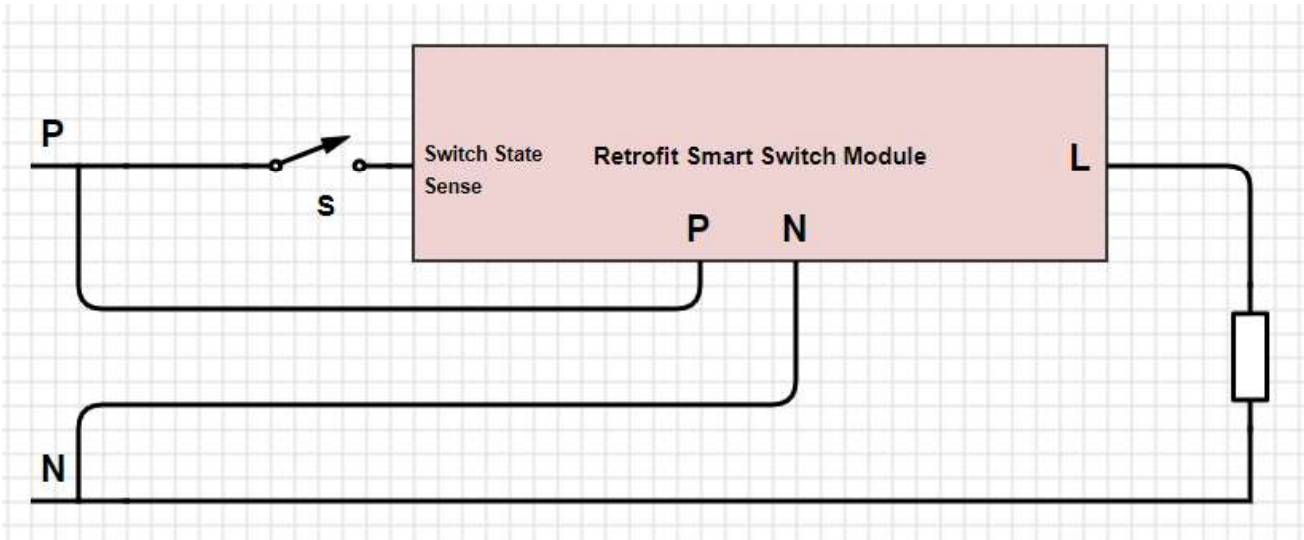
Aim

1. To wire up the circuit of a single load controlled by one SPST switch.
2. To make the conventional SPST switch circuit smart using retrofit smart switch module.
3. To wire up the circuit of a two switch circuit with a smart switch module.

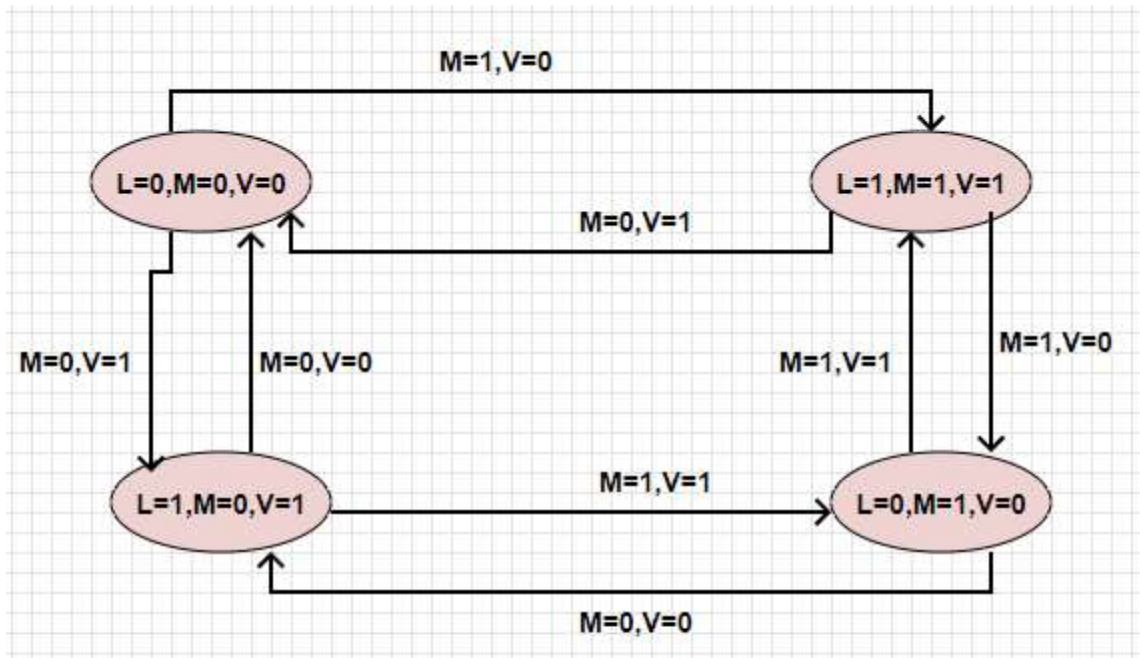
Procedure

1. Circuit of one SPST Switch circuit controlling a single load is wired & tested.
2. Using a Suitable Retrofit Smart Switch Module, connections of the conventional SPST switch circuit are modified as shown below.
3. Develop the functional state diagram of the system for exercising ON-OFF Control of the system.

Wired Connections of SPST switch circuit with retrofit smart switch module



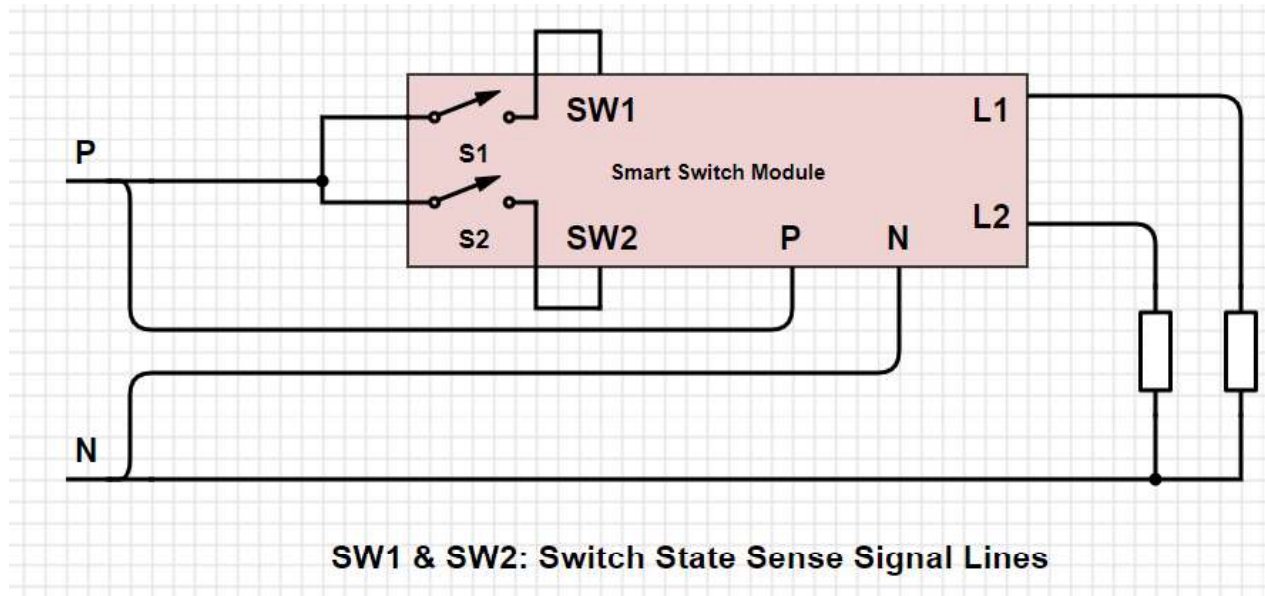
Functional State Diagram of the System for Exercising ON-OFF Control of the System



Smart Switch Module Circuit

1. Connections are arranged as shown in the figure below.
2. Testify both manual & Mobile App Control Modes of the circuit.

Wired Connections of the Smart Switch Module



Result

The experiment has been completed successfully.